

MATH 1564 — Notes 09/08

Matrices

$$A = [T(e_1) \ \cdots \ T(e_n)]$$

A has n columns

A has m rows

$$Ax = b$$

$$\text{span}\{T(e_1), \dots, T(e_n)\}$$

$$= \text{span}\{\text{columns of } A\}$$

$$= \{Ax : x \text{ in } \mathbb{R}^n\}$$

$$= \{T(x) : x \text{ in } \mathbb{R}^n\}$$

A has a pivot in each row

A has a pivot in each column

Linear maps

$$T(x) = Ax$$

Domain of $T = \mathbb{R}^n$

Codomain of $T = \mathbb{R}^m$

$$T(x) = b$$

Image of T

onto
one-to-one

Theorem 1: The composition of linear maps is linear.

Proof: $\mathbb{R}^n \xrightarrow{T} \mathbb{R}^m \xrightarrow{S} \mathbb{R}^k$.

A map f is linear $\iff f(c_1v_1 + c_2v_2) = c_1f(v_1) + c_2f(v_2)$.

T is linear:

$$\begin{aligned} S(T(c_1v_1 + c_2v_2)) &= S(c_1T(v_1) + c_2T(v_2)) \\ &= c_1S(T(v_1)) + c_2S(T(v_2)). \end{aligned}$$

Let A, B be matrices with $T(x) = Ax$ and $S(y) = By$. What are the sizes of A, B ?

$$\begin{aligned} \mathbb{R}^n &\xrightarrow{T} \mathbb{R}^m \xrightarrow{S} \mathbb{R}^k \\ A : m \times n, &\quad B : k \times m. \end{aligned}$$

Definition: Product of matrices. The product BA is the standard matrix of $S \circ T$.

What is the size of BA ? $k \times n$.

Does AB make sense? Yes if $k = n$. No if $k \neq n$.

What is the standard matrix of the map $T(x) = x$ for all x ?

$$T(x) = Ax = \begin{bmatrix} 1 & & 0 \\ & \ddots & \\ 0 & & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}.$$

Example: $A_{m \times n} I_n = A$, since $S \circ T = S$ when $T(x) = x$.

Example: $I_m A_{m \times n} = A$. ($I_1 = [1]$.)

Example: Is $A(BC) = (AB)C$? (Associativity.)

$$S \circ R \circ T = S(R(T(x))). \quad \text{Yes.}$$

Example: Is $AB = BA$? Not commutative.

Formula for product of matrices

Theorem 2: If $A = [a_1 \cdots a_n]$ and B is $k \times m$, then $BA = [Ba_1 \cdots Ba_n]$.

Example:

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 5 & 0 \\ 4 & 2 \end{bmatrix} = \begin{bmatrix} 13 & 4 \\ 31 & 8 \end{bmatrix}.$$

Proof: The standard matrix for $S \circ T$ is

$$[S(T(e_1)) \cdots S(T(e_n))] = [BT(e_1) \cdots BT(e_n)],$$

where $S(y) = By$, $T(x) = Ax$. The matrix for T is $[T(e_1) \cdots T(e_n)] = A$, with $T(e_1) = a_1, \dots, T(e_n) = a_n$.

Notation: a_{ij} = entry of A at row i , column j .

Example: $A = \begin{bmatrix} 1 & -1 & 0 \\ 7 & 3 & -2 \end{bmatrix}$

$$a_{23} = -2, \quad a_{12} = -1, \quad a_{31} = \text{undefined}.$$

Theorem 3: If $A = (a_{ij})$ is $m \times n$ and $B = (b_{ij})$ is $n \times k$, then the ij entry of AB is

$$a_{i1}b_{1j} + \cdots + a_{in}b_{nj}.$$

Proof: The j th column of AB is

$$A \begin{bmatrix} b_{1j} \\ \vdots \\ b_{nj} \end{bmatrix} = b_{1j} \begin{bmatrix} a_{11} \\ \vdots \\ a_{m1} \end{bmatrix} + \cdots + b_{nj} \begin{bmatrix} a_{1n} \\ \vdots \\ a_{mn} \end{bmatrix}.$$

Example: The 21 entry of

$$\begin{bmatrix} 1 & 2 \\ 3 & -1 \end{bmatrix} \begin{bmatrix} -1 & 0 \\ 1 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 6 \\ -4 & -3 \end{bmatrix}$$

is $\boxed{-4}$.

Example: If A, B are $n \times n$ matrices, do we expect $AB = BA$?

$$\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \quad \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}. \quad \text{No.}$$

If $AB = 0$, we can't expect $A = 0$ or $B = 0$.

If $AB = AC$, $A \neq 0$, we can't expect $B = C$.

$$A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} : \quad \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 3 & 4 \\ 0 & 0 \end{bmatrix}, \quad \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 5 & 6 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 3 & 4 \\ 0 & 0 \end{bmatrix}.$$

Example:

$$\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}.$$

Example (powers). $A = \begin{bmatrix} 2 & 0 \\ 0 & -1 \end{bmatrix}$ (a **diagonal matrix**):

$$A^2 = \begin{bmatrix} 2 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & -1 \end{bmatrix} = \begin{bmatrix} 2^2 & 0 \\ 0 & (-1)^2 \end{bmatrix} = \begin{bmatrix} 4 & 0 \\ 0 & 1 \end{bmatrix}.$$

Theorem 4: $A^n = \begin{bmatrix} 2^n & 0 \\ 0 & (-1)^n \end{bmatrix}$.

Proof (induction): $n = 1$: true.

Hypothesis: $A^{n-1} = \begin{bmatrix} 2^{n-1} & 0 \\ 0 & (-1)^{n-1} \end{bmatrix}$.

Prove:

$$A^n = A^{n-1}A = \begin{bmatrix} 2^{n-1} & 0 \\ 0 & (-1)^{n-1} \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & -1 \end{bmatrix} = \begin{bmatrix} 2^n & 0 \\ 0 & (-1)^n \end{bmatrix}.$$

Similarly, for $k > 0$:

$$\begin{bmatrix} d_1 & & 0 \\ & \ddots & \\ 0 & & d_n \end{bmatrix}^k = \begin{bmatrix} d_1^k & & 0 \\ & \ddots & \\ 0 & & d_n^k \end{bmatrix}.$$

Example: $\begin{bmatrix} \pm 1 & 0 \\ 0 & \pm 1 \end{bmatrix}^2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$.

Scaling matrices.

$$c \begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \cdots & a_{mn} \end{bmatrix} = \begin{bmatrix} ca_{11} & \cdots & ca_{1n} \\ \vdots & \ddots & \vdots \\ ca_{m1} & \cdots & ca_{mn} \end{bmatrix}.$$

Example: $2 \begin{bmatrix} 1 & 3 \\ -1 & 5 \end{bmatrix} = \begin{bmatrix} 2 & 6 \\ -2 & 10 \end{bmatrix}.$

Adding matrices of the same size.

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} + \begin{bmatrix} -1 & 5 \\ 0 & 6 \end{bmatrix} = \begin{bmatrix} 1-1 & 2+5 \\ 3+0 & 4+6 \end{bmatrix} = \begin{bmatrix} 0 & 7 \\ 3 & 10 \end{bmatrix}.$$

Recall vectors can be scaled and added; matrices of the same size can be thought of the same way.

Properties. For matrices A, B, C of compatible sizes and scalars r, s :

$$\begin{aligned}A + 0 &= 0 + A = A \\(A + B) + C &= A + (B + C) \\r(A + B) &= rA + rB \\(r + s)A &= rA + sA \\(rs)A &= r(sA) \\A(B + C) &= AB + AC \\(A + B)C &= AC + BC\end{aligned}$$

Example: For A, B both $n \times n$:

$$(A + B)^2 = (A + B)(A + B) = A^2 + AB + BA + B^2.$$

Example:

$$(A + B)(A - B) = A^2 - AB + BA - B^2.$$

(Note: since $AB \neq BA$ in general, these do *not* simplify to $A^2 + 2AB + B^2$ or $A^2 - B^2$ as they would for numbers.)